Ecology: abundance and distribution

9C Threat from disease · Lesson 6 of 7

Date _____ Class _____

Today I can

- ☐ I can choose suitable apparatus for measuring the abundance and distribution of organisms.
- ☐ I can use data from a sample to estimate a population size and say how accurate the estimate is.

What happens today

1. Do Now: counting the uncountable
2. Learn: Abundance and sampling · Estimating population size · How accurate is the estimate? · and more
3. Practical: Quadrat sampling
4. Challenge, then the exit ticket

Student Book pages 48 to 49 · Workbook pages 34 to 35

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
abundance	how common a species is in a habitat
distribution	how organisms are spread out in an area
population	all the organisms of one species in an area
sample	a small selected part of the total number
estimate	an approximate calculation of something
quadrat	a square frame used to sample plants or slow-moving animals
transect	a line along which samples are taken
pitfall trap	a container sunk into the ground to catch small ground animals
sweep net	a net swept through long grass to catch insects
random	chosen by chance, not by the person
biodiversity	the number of different species in an area

Quick Review

counting the uncountable

During a disease outbreak, scientists cannot count every infected person, so they estimate.

1 What does 'estimate' mean?

2 Suggest how you could estimate the number of dandelions on the school field without counting them all.

3 Name one abiotic and one biotic factor that could affect how many dandelions there are.

Core 1 of 5

Everyone does these. Answer on the card or under it.

1 Match each word to its definition. Draw a line.

abundance	an approximate calculation of something
distribution	the number of different species in an area
sample	how common a species is in a habitat
estimate	a small selected part of the total number
biodiversity	how organisms are spread out in an area

2 Tick one box in each row to show the best method for sampling each organism.

Organism	pitfall trap	sweep net	quadrat	pooter
ground beetles				
insects in long grass				
daisies on a lawn				
tiny insects on a leaf				
limpets on a rocky shore				

Core 2 of 5

Everyone does these. Answer on the card or under it.

3 Students counted one species of snail in eight quadrats placed at random in a field 30 m by 25 m. Each quadrat was 0.5 m by 0.5 m.

Quadrat	1	2	3	4	5	6	7	8
Number of snails	3	1	0	2	4	1	2	3

a) Calculate the total number of snails counted.

b) Calculate the total area sampled by the eight quadrats, in m2.

Core 3 of 5

Everyone does these. Answer on the card or under it.

3 *continued*

c) Calculate the area of the whole field, in m2.

d) Estimate the population of this snail species in the whole field. Show your working.

Core 4 of 5

Everyone does these. Answer on the card or under it.

4 Record your own quadrat results, then estimate the population of your plant in the 10 m by 10 m area.

a) Complete the table.

Quadrat	Coordinates	Number of plants
1		
2		
3		
4		
5		
6		
7		
8		
total		

Core 5 of 5

Everyone does these. Answer on the card or under it.

4 *continued*

b) Estimate the population: total x 100 m2 / 2 m2. Show your working.

Challenge 1 of 2

When Core is finished. Full sentences.

C1 Ecologists compared the biodiversity of birds in rainforest and in oil palm plantations that replaced some of the forest. They walked along transects through each area, recording every bird they saw or heard.

a) Suggest why the scientists used transects rather than quadrats.

b) Give two things the scientists should keep the same so that the results from the two areas can be compared fairly.

Challenge 2 of 2

When Core is finished. Full sentences.

C1 *continued*

c) Oil palm plantations have trees planted in rows, all the same age; rainforest has many kinds of trees of different sizes. Predict in which area the birds would show a more regular distribution, and give a reason.

C2 There are many types of grass, each with slightly different leaf shapes. Describe how you would find out which of four species was the most abundant in a large field.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Why do ecologists take samples instead of counting every organism?

2 5 daisies were found in one 0.25 m2 quadrat. The lawn is 40 m2. Estimate the daisy population.

3 Give one reason why quadrats should be placed at random.

Today I can: Goal 1 ☐ Goal 2 ☐